Theory and manual

for

AlkCalc

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I. Introduction

Bound single-electron eigenfunctions of alkali-metal atoms and alkaline-earth metal ions are of relevance in quantum computing, quantum information processing, quantum simulation, Rydberg physics, quantum optics, ... However, expect for Hydrogen, it is not possible to solve the time-independent Schrödinger equation analytically. Because of this, there is the need to obtain eigenenergies, i.e., atomic spectra, and eigenstates numerically. The library `AlkCalc` is intended for exactly this purpose. In the various README files included in `AlkCalc` technical details for running the software are provided. However, in these README files, there is no theoretical discussion on the theory behind the code. This is the purpose of the present document.

In this document the class of Hamiltonians which `AlkCalc` is designed to diagonalise is presented. Moreover, this document formally defines each quantity which `AlkCalc` is designed to compute. Therefore, this document serves as a reference for the user. In particular, the final section is a complete reference manual for the library functions of `AlkCalc` .

II. Theory

We start by considering a single atom/ion. Subsequently, we describe the finite element method (FEM) used to numerically solve the Schrödinger equation. Finally, we describe how `AlkCalc` computes radial matrix elements, oscillator strengths, and state lifetimes.

i. Single-atom/ion theory

The starting point is an alkali-metal atom or alkaline-earth-metal ion. We are interested in systems where all electrons are in closed shells except for one (e.g., ^{1}H, ^{38}Rb, ^{88}Sr^{+}). This allows a Kepler-problem description for the 'effective nucleus' (the nucleus 'screened' by the closed-shell electrons) and the valence electron. To be concrete, we study the Hamiltonian

$$H = \frac{\boldsymbol{p}^2}{2m} + V(\|\boldsymbol{r}\|\,; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,, \tag{II.1}$$

where $m = Mm_\mathrm{e}/(M + m_\mathrm{e})$ is the reduced mass of the system, composed of the electron's mass m_e and the atom's or ion's mass M without the valence electron. The centre-of-mass motion is decoupled, such that $\boldsymbol{r}, \boldsymbol{p}$ are the relative canonical co-ordinates, i.e., $\boldsymbol{r}$ is the distance vector between the centre-of-mass position of the effective nucleus and the valence electron. Moreover, $\boldsymbol{L}$, $\boldsymbol{S}$, and $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$ are the orbital angular momentum operator, the electron's spin operator, and the total angular momentum operator, respectively. The potential V is not a simple Coulomb potential, but a generalised potential which includes different contributions (these arise when one starts with the full many-body electron Hamiltonian and incorporates relativistic and quantum electrodynamics effects). Following Refs. [1–3] we write:

$$V(\|\boldsymbol{r}\|\,; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = V_\mathrm{C}(\boldsymbol{r}) + V_\mathrm{P}(\boldsymbol{r}) + V_\mathrm{R}(\boldsymbol{r}; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.2}$$

The first contribution is

$$V_\mathrm{C}(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{Z_\mathrm{n}(r)}{r}\,,\; Z_\mathrm{n}(r) = Z_\mathrm{c} + (Z - Z_\mathrm{c})\mathrm{e}^{-k_1 r} + k_2 r \mathrm{e}^{-k_3 r} - k_4 r^2 \mathrm{e}^{-k_3 r}\,. \tag{II.3}$$

This is a modified Coulomb potential, adjusted for small radii due to the effective nucleus. Here, Z is the nuclear charge, Z_c the nuclear charge of the effective nucleus, and the k_j $(j = 1, 2, 3, 4)$ are empirically determined fitting parameters; see Refs. [1–3]. The second contribution is

$$V_{\mathrm{P}}(r) = -\frac{e^2}{(4\pi\varepsilon_0)^2}\frac{\alpha_{\mathrm{d}}}{2r^4}\left[1 - \exp\left(-\frac{r^6}{r_{\mathrm{c}}^6}\right)\right], \tag{II.4}$$

which accounts for the polarisation of the screened nucleus due to the presence of the valence electron. Here, r_c is the effective nucleus' size and α_d its polarisability. Finally,

$$V_{\mathrm{R}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{1}{2m^2c^2}\frac{V'_{\mathrm{NR}}(r)}{rN(r)}\boldsymbol{L}\cdot\boldsymbol{S}, \; N(r) = \left[1 - \frac{V_{\mathrm{NR}}(r)}{2mc^2}\right]^2, \; V_{\mathrm{NR}}(r) = V_{\mathrm{C}}(r) + V_{\mathrm{P}}(r), \tag{II.5}$$

which describes (relativistic) coupling between the orbital angular momentum and the valence electrons' spin. In this expression, V_{NR} is the non-relativistic model potential and c the speed of light. In Refs. [1, 3] this potential is used for ions. We note that in Ref. [4] the simplified version

$$V_{\mathrm{R}}(r) = \frac{\alpha}{2r^3}\boldsymbol{L}\cdot\boldsymbol{S} \tag{II.6}$$

is employed for atoms. Here, α is the fine-structure constant. However, we use the first, more precise, version for both ions and atoms. The spin-operator $\boldsymbol{S}$ in Eqs. (II.6, II.5) is conveniently expressed as

$$\boldsymbol{S} = \frac{\hbar}{2}\begin{bmatrix}\sigma_x \\ \sigma_y \\ \sigma_z\end{bmatrix}, \tag{II.7}$$

with the Pauli matrices

$$\sigma_x = \begin{bmatrix}0 & 1 \\ 1 & 0\end{bmatrix}, \; \sigma_y = \begin{bmatrix}0 & -\mathrm{i} \\ \mathrm{i} & 0\end{bmatrix}, \; \sigma_z = \begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix}, \tag{II.8}$$

which is possible because we deal with a single electron, i.e., a particle with spin $1/2$. Moreover, we rewrite the product $\boldsymbol{L}\cdot\boldsymbol{S}$ in terms of the total angular momentum operator $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}\left[\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2\right]. \tag{II.9}$$

For further simplification the square of the momentum operator is expressed in spherical coordinates,

$$\boldsymbol{p}^2 = (-\mathrm{i}\hbar\nabla)^2 = -\hbar^2\nabla^2 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\boldsymbol{L}^2}{r^2}. \tag{II.10}$$

The second contribution is the standard centrifugal energy contribution, analogously to the Kepler problem. With these simplification we rewrite the Hamiltonian as

$$H = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2), \; V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{\boldsymbol{L}^2}{2mr^2} + V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2). \tag{II.11}$$

The full Hilbert space this Hamiltonian acts on is split into an angular momentum factor and a radial one. By construction, a basis of the angular momentum Hilbert space is given by the states $|l, s, j, m_j\rangle$, which are eigenstates of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z, i.e., constitute the coupled basis, also called 'fine-structure' basis. Note that it must be possible to construct an eigenbasis of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z simultaneously, because they commute pairwise. In particular, these states are constructed as linear combinations of the uncoupled basis states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$, employing Clebsch-Gordan coefficients. Here, $Y_{l,m_l} : [0, \pi] \times [0, 2\pi] \to \mathbb{C}$ are spherical harmonics defined as (the so-called 'Condon-Shortley phase' we include in the definition of the Legendre polynomials below, i.e., the factor $(-1)^{m_l}$ is *not* missing here; see e.g. Ref. [5])

$$Y_{l,m_l}(\vartheta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m_l)!}{(l+m_l)!}} P_l^{m_l}(\cos(\vartheta))\, \mathrm{e}^{\mathrm{i} m_l \varphi} \,, \tag{II.12}$$

where the associated Legendre polynomials $P_l^{m_l} : [-1, 1] \to \mathbb{R}$ are defined for $l = 0, 1, 2, \ldots$ and $m_l = 0, 1, \ldots, l-1, l$, via the recursion relation

$$(l - m_l) P_l^{m_l}(x) = x(2l-1) P_{l-1}^{m_l}(x) - (l + m_l - 1) P_{l-2}^{m_l}(x) \,, \tag{II.13}$$

with starting point

$$P_{m_l}^{m_l}(x) = (-1)^{m_l} \frac{(2m_l)!}{2^{m_l} m_l!} (1 - x^2)^{m_l/2} \,, \tag{II.14}$$

in combination with $P_l^{m_l}(x) = 0$ for $l < m_l$. Negative values of m_l are treated with the relation

$$P_l^{-m_l} = (-1)^{m_l} \frac{(l - m_l)!}{(l + m_l)!} P_l^{m_l} \,. \tag{II.15}$$

Moreover, $\boldsymbol{\chi}_{s,m_s}$ are two-spinors defined as

$$\boldsymbol{\chi}_{\frac{1}{2},\frac{1}{2}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \boldsymbol{\chi}_{\frac{1}{2},-\frac{1}{2}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{II.16}$$

Explicitly (for the case of interest $s = 1/2$), we find the following expressions for the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m_j}$:

$$\boldsymbol{\Phi}_{l,s,j,m_j} = \langle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s} + \langle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s} \,, \tag{II.17}$$

where the scalar products are (possibly vanishing) Clebsch-Gordan coefficients. We further find that these states transform according to the irreducible representations of the group SU(2) under the action of the SU(2)-representation $\rho(\boldsymbol{\alpha}) = \exp(\mathrm{i}\boldsymbol{\alpha} \cdot \boldsymbol{J})$, where $\boldsymbol{\alpha} \in \mathbb{R}^3$ is a vector of length $\|\boldsymbol{\alpha}\| \in [0, 2\pi)$. Moreover, the angular momentum Hamiltonian is invariant under the action of ρ, such that it is automatically block diagonal. The irreducible representations of SU(2) are $(2j+1)$-dimensional, with $j = 0, 1/2, 1, 3/2, 2, \ldots$, such that the blocks have dimensions $\mu_j(2j+1)$, where μ_j is the multiplicity of the irreducible representation j in the direct sum decomposition of ρ. The eigenvalues associated to the blocks are operators on the radial Hilbert space. The multiplicities are $\mu_j = 2$ and $\mu_0 = 0$, because for each fixed j there are two possibilities: $l = j - 1/2$ and $l = j + 1/2$. Explicitly, the Hamiltonian is written as

$$\bigoplus_{m_j=-j}^{j} \langle \boldsymbol{\Phi}_{l,s,j,m_j}, H \boldsymbol{\Phi}_{l,s,j,m_j} \rangle \equiv \bigoplus_{m_j=-j}^{j} H_{l,s,j} \,, \tag{II.18}$$

where (with the arbitrary choice $m_j = j$) we defined

$$H_{l,s,j} = \langle \Phi_{l,s,j,j}, H\Phi_{l,s,j,j} \rangle = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\text{eff}}(r; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1]) . \quad \text{(II.19)}$$

We note that the fact that the matrix element does not depend on m_j is a consequence of the SU(2)-symmetry of the Hamiltonian. Therefore, exactly this is the reason for the energies being degenerate in the total magnetic angular momentum quantum number m_j[1] The symmetry consideration up until now simplified the problem to the block associated to the irreducible representation of SU(2). What is left to solve are the eigenproblems

$$H_{l,s,j}R_{n,l,s,j} = E_{n,l,s,j}R_{n,l,s,j} , \quad \text{(II.20)}$$

where $R_{n,l,s}$ is the n-th (radial) eigenstate of $H_{l,s,j}$. Therefore, if we manage to obtain $R_{n,l,s,j}$ the full spectrum is known. Of course, resonances (states associated to the channel $\mathbf{\Phi}_{l,s,j,m_j}$ which are not bound) exist, but in the following we will focus on bound states.

ii. Numerical method

The radial eigenstates, generally, cannot be obtained analytically. `AlkCalc` provides a platform to solve for these states numerically. To make the problem addressable by a computer, we express the eigenproblem in terms of dimensionless quantities. To this end we express the Hamiltonians $H_{l,s,j}$ in units of Hartree, $E_\text{H} = 1/(4\pi\varepsilon_0 a_\text{B})$, where $a_\text{B} = \hbar/(m_\text{e}c\alpha) \approx 52.9\,\text{pm}$ is Bohr's radius. The radial eigenfunctions we parametrise as

$$R_{n,l,s,j}(r) = \frac{1}{\sqrt{a_\text{B}^3}}\frac{f_{n,l,s,j}(r/a_\text{B})}{r/a_\text{B}} . \quad \text{(II.21)}$$

where $f_{n,l,s,j}$ is dimensionless. The eigenproblem in Eq. (II.20) in the new eigenfunctions takes the form

$$\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \frac{l(l+1)}{2t^2} + C\frac{V(a_\text{B}t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_\text{H}}\right] f_{n,l,s,j}(t) = \bar{E}_{n,l,s,j} f_{n,l,s,j}(t) , \quad \text{(II.22)}$$

where we defined

$$C = \frac{1}{1 + m_\text{e}/M} \quad \text{(II.23)}$$

and the eigenenergies may be extracted from the relation

$$\frac{E_{l,s,j,n}}{E_\text{H}} = \frac{\bar{E}_{n,l,s,j}}{C} . \quad \text{(II.24)}$$

1 Generally, there is no reason that the blocks $H_{l,s,j}$ are separate blocks for $l = j+1/2$ and $l = j-1/2$, because they correspond to the same irreducible representation j. However, the states we choose are already chosen in a way such that the blocks are already separated. In this case we know how we should choose the states already from the fact that the Hamiltonian explicitly depends on $\boldsymbol{L}$: if it would not be so obvious, we would not immediately know what the multiplicity index l (and s) should physically be to sort the eigenvalues into separate blocks.

Rewriting the normalisation condition in terms of the eigenfunctions $f_{n,l,s,j}$ leads to

$$1 = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}^2(r) = \int_0^\infty (a_\mathrm{B}\,\mathrm{d}t)(a_\mathrm{B}t)^2 \frac{1}{a_\mathrm{B}^3} \frac{f_{n,l,s,j}^2(t)}{t^2} = \int_0^\infty \mathrm{d}t\, f_{n,l,s,j}^2(t)\,. \tag{II.25}$$

Here we assumed that the (radial) eigenfunctions are real-valued, which is justified as the Hamiltonian is not only an hermitian but also a symmetric operator. In conclusion, computing the normalised eigenfunctions $f_{n,l,s,j}$ yields the (automatically normalised) radial eigenstates by employing Eq. (II.21) for the conversion.

To solve the eigenproblem in Eq. (II.22) numerically, we employ a FEM instead of the conceptually (and programmatically) more simple finite difference method (FDM) for, mainly[2], one reason: With a FEM we are completely free in choosing the mesh (the position of the discretisation points) for the radial domain, while keeping the associated finite-dimensional matrix problem real and symmetric. This is important because the problem we face presents itself with drastically different energy scales, introduced by the diverging potential: the eigenfunctions $f_{n,l,s,j}$ oscillate quickly for small arguments and slowly for large ones, where the valence electron is almost free. Thus, a mesh tailored to this behaviour leads to a significant reduction of computational resources (the fast oscillation can be efficiently resolved without introducing 'too many' points in the slowly oscillating regime).

The first step in setting-up the FEM is to agree on a finite closed interval $I = [0, t_\mathrm{tmax}]$, where $t_\mathrm{max} > 0$ sets the maximal value of the domain we consider. The value t_max must be chosen large enough to ensure that the boundary condition $f_{n,l,s,j}(t \geq t_\mathrm{max}) = 0$ is fulfilled. This equal sign is actually never true, because the radial eigenstates asymptotically approach zero. However, if we reach zero up to machine precision, it is numerically true. Of course, one must ensure that the finite set of solutions that shall be computed has support within this interval. The mesh we define as a family $(t_k)_{k=1,\dots,N-1}$ in I with the properties $t_0 = 0$, $t_{N-1} = t_\mathrm{max}$, and $t_0 < t_1 < \cdots < t_{N-2} < t_{N-1}$, where we say $N \geq 3$. Of course, typical values are $N \sim 10^5$-10^7. Next, we define the derived family of step sizes $(h_k)_{k=1,\dots,N-1}$, $h_k = t_k - t_{k-1}$. The elements $(\phi_k)_{k=0,\dots,N-1}$, $\phi_k : I \to [0,1]$, we define as

$$\phi_k(t) = \begin{cases} (t - t_{k-1})/(t_k - t_{k-1})\,, & t \in (t_{k-1}, t_k) \\ (t_{k+1} - t)/(t_{k+1} - t_k)\,, & t \in (t_k, t_{k+1}) \\ 0\,, & \text{else} \end{cases}\ . \tag{II.26}$$

for $k = 1, \dots, N-2$, and for $k = 0$ and $k = N-1$ the Dirichlet boundary conditions $f_{n,l,s,j}(0) = f_{n,l,s,j}(t_\mathrm{max}) = 0$ lead to the canonical choice $\phi_0 = \phi_{N-1} \equiv 0$. To obtain a matrix problem which approximates Eq. (II.22) we compute projections of it onto the elements with the real scalar product ('dummy' functions $\alpha, \beta : I \to \mathbb{R}$)

$$B(\alpha, \beta) = \int_I \mathrm{d}t\, \alpha(t)\beta(t)\,. \tag{II.27}$$

2 It also helps to save some disk space, because one must store much less data to represent the eigenfunctions $f_{n,l,s,j}$. Typically, disk space is not a relevant factor these days, but we see no need to be unnecessarily wasteful with it either.

To simplify the notation we fix the quantum numbers n, l, s, and j and say $f \equiv f_{n,l,s,j}$. Moreover, we collect all potential contributions into the new potential v, defined as

$$v(t) = \frac{l(l+1)}{2t^2} + C\frac{V(a_\mathrm{B}t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_\mathrm{H}} . \tag{II.28}$$

Finally, we say $\lambda = \bar{E}_{n,l,s,j}$, such that Eq. (II.22) simply becomes

$$-\frac{1}{2}f''(t) + v(t)f(t) = \lambda f(t) . \tag{II.29}$$

Before we project this equation onto the elements, we approximately expand f in them:

$$f = \sum_{k=1}^{N-2} f_k \phi_k . \tag{II.30}$$

This expansion automatically fulfils the Dirichlet boundary conditions $f(0) = f(t_\mathrm{max}) = 0$. Projection, i.e., applying the map $B(\phi_k, \cdot)$ for all $k = 1, \dots, N-2$, yields a matrix eigenproblem whose solutions converge to solutions of Eq. (II.22) as N tends to infinity:

$$(K + V)\boldsymbol{f} = \lambda M \boldsymbol{f} . \tag{II.31}$$

In this expression $M_{kk'} = B(\phi_k, \phi_{k'})$ is the so-called mass matrix, $K_{kk'} = B(\phi'_k, \phi'_{k'})/2$ the so-called stiffness matrix[3], and $V_{kk'} = B(\phi_k, v\phi_{k'})$ the potential energy contribution. Each matrix is $(N-2)$-dimensional, since $k, k' = 1, \dots, N-2$, and M and K can be computed analytically; in particular

$$M = \frac{1}{6}\begin{bmatrix} 2(h_1+h_2) & h_2 & & & \\ h_2 & 2(h_2+h_3) & h_3 & & \\ & h_3 & 2(h_3+h_4) & h_4 & \\ & & \ddots & \ddots & h_{N-2} \\ & & & h_{N-2} & 2(h_{N-2}+h_{N-1}) \end{bmatrix} \tag{II.32}$$

and

$$K = \frac{1}{2}\begin{bmatrix} \frac{1}{h_1}+\frac{1}{h_2} & -\frac{1}{h_2} & & & \\ -\frac{1}{h_2} & \frac{1}{h_2}+\frac{1}{h_3} & -\frac{1}{h_3} & & \\ & -\frac{1}{h_3} & \frac{1}{h_3}+\frac{1}{h_4} & -\frac{1}{h_4} & \\ & & \ddots & \ddots & -\frac{1}{h_{N-2}} \\ & & & -\frac{1}{h_{N-2}} & \frac{1}{h_{N-2}}+\frac{1}{h_{N-1}} \end{bmatrix} . \tag{II.33}$$

Note that both of these matrices are symmetric. The potential matrix, generally, must be computed numerically. However, here we choose the simple approximation

$$B(\phi_k, v\phi_{k'}) \approx \frac{v(t_k) + v(t_{k'})}{2} M_{kk'} , \tag{II.34}$$

3 Note that because the elements obey the Dirichlet boundary conditions $\phi_k(0) = \phi_k(t_\mathrm{max}) = 0$, $k = 1, \dots, N-2$, integrating by parts allows to write $B(\phi''_k, \phi_{k'}) = -B(\phi'_k, \phi'_{k'})$.

which is justified if the step size is small enough. Note that this particular choice of approximations keeps the matrix symmetric, such that $K + V$ is symmetric. Furthermore, M represents a scalar product and is thus positive definite. This means that the matrix eigenproblem in Eq. (II.31) can be efficiently solved with the Lanczos algorithm.

Next, we express the normalisation condition in Eq. (II.25) in terms of our elements and the vector $\boldsymbol{f}$. To this end we calculate

$$\int_0^\infty \mathrm{d}t\, f^2(t) \approx \int_I \mathrm{d}t\, f^2(t) = \sum_{k=1}^{N-2}\sum_{k'=1}^{N-2} f_k f_{k'} B(\phi_k, \phi_{k'}) = \boldsymbol{f}^{\mathrm{T}} M \boldsymbol{f} = \|\boldsymbol{f}\|_M^2 \stackrel{!}{=} 1\,, \tag{II.35}$$

i.e., the length of $\boldsymbol{f}$ must be unity in the by the scalar product M induced norm $\|\cdot\|_M$.

iii. Radial matrix elements

To compute radial matrix elements, we first calculate, for $p \in \mathbb{R}$ such that the integral exists,

$$\begin{aligned}\frac{r_p^{n,l,s,j;n',l',s',j'}}{a_{\mathrm{B}}^p} &\equiv \frac{1}{a_{\mathrm{B}}^p}\langle R_{n,l,s,j}, \mathrm{id}^p R_{n',l',s',j'}\rangle = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}(r)\frac{r^p}{a_{\mathrm{B}}^p} R_{n',l',s',j'}(r) \\ &= \int_0^\infty (\mathrm{d}t\, a_{\mathrm{B}})(a_{\mathrm{B}}t)^2 f_{n,l,s,j}(t)\frac{(a_{\mathrm{B}}t)^p}{a_{\mathrm{B}}^p}\frac{1}{a_{\mathrm{B}}^3}\frac{f_{n',l',s',j'}(t)}{t^2} = B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'})\,.\end{aligned} \tag{II.36}$$

Here, 'id' denotes the identity on the non-negative real numbers. Next, we approximate this result with solutions to the matrix eigenproblem in Eq. (II.31), i.e., we represent $f_{n,l,s,j}$ and $f_{n',l',s',j'}$ approximately with the vectors $\boldsymbol{f}_{n,l,s,j}$ and $\boldsymbol{f}_{n',l',s',j'}$, respectively [see Eq. (II.30)]. This allows to write

$$B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'}) \approx \boldsymbol{f}_{n,l,s,j}^{\mathrm{T}} R_p \boldsymbol{f}_{n',l',s',j'}\,, \tag{II.37}$$

where the components of the real symmetric matrix R_p are defined as

$$(R_p)_{k,k'} = B(\phi_k, \mathrm{id}^p \phi_{k'}) = \begin{cases} B(\phi_k, \mathrm{id}^p \phi_{k+1}), & k' = k+1 \text{ or } k = k'+1\ (0 < k, k' < N-2) \\ B(\phi_k, \mathrm{id}^p \phi_k), & k' = k\ (0 < k, k' \leq N-2) \\ 0, & \text{else} \end{cases}\,. \tag{II.38}$$

To numerically compute the remaining integrals we use a two-point Gaußian quadrature. This method approximates the integral as [6]:

$$\int_a^b \mathrm{d}t\, \alpha(t) \approx \frac{b-a}{2}\left[\alpha\left(\frac{a+b}{2} - \frac{b-a}{2\sqrt{3}}\right) + \alpha\left(\frac{a+b}{2} + \frac{b-a}{2\sqrt{3}}\right)\right], \tag{II.39}$$

for any sufficiently smooth function α and $a < b$. Note that this result is exact if α is a polynomial of degree three, which is the case for radial dipole matrix elements. This method is straightforward to implement in a computer program, such that the problem of calculating matrix elements is effectively reduced to computing matrix products. However, note that for too large p there can be numerical overflow or underflow when computing t^p. But, for 64-bit floating point arithmetic, $p = \pm 10$ is easily possible (if the integral exists), such that, typically, all practical cases are included. Explicitly, we could verify that for Hydrogen the range $p = -2, \ldots, 10$ yields reasonably good results. Finally, we remark that one can improve on the precision by employing higher-order Gaußian quadrature algorithms as described in [6].

iv. Oscillator strengths

The so-called 'oscillator strength', for a transition from an initial state $\boldsymbol{\Psi}_\mathrm{i} \equiv \boldsymbol{\Psi}_{n',l',s,j',m_j'}$ to a final state $\boldsymbol{\Psi}_\mathrm{f} \equiv \boldsymbol{\Psi}_{n,l,s,j,m_j}$ is [5]

$$f_{\mathrm{i}\to\mathrm{f}} = \sum_{m_j'=-j'}^{j'} \frac{2}{3}\bar{E}_{\mathrm{f,i}} |\langle \boldsymbol{\Psi}_\mathrm{f}, \boldsymbol{r}/a_\mathrm{B}\boldsymbol{\Psi}_\mathrm{i}\rangle|^2 \,, \tag{II.40}$$

with the energy difference $E_\mathrm{H}\bar{E}_{\mathrm{f,i}} = E_\mathrm{f} - E_\mathrm{i}$. Note that we sum over all final m_j', because we are interested in transitions between the fine-structure states without resolving the Zeeman sublevels. The oscillator strength is a measure for how strongly the induced Hertzian (electric) dipole oscillates: the radiation power is large for spatially large dipoles and fast oscillation frequencies, captured by the matrix element and energy difference, respectively. In principle, we already have all the machinery to numerically compute this quantity, however, one can reduce the expression to a much simpler form by using angular momentum algebra, in particular, relations from [7]. To simplify the expression, we first use (Condon-Shortley phase)

$$r_x = \sqrt{\frac{2\pi}{3}} r(Y_{1,-1} - Y_{1,1})\,,\ r_y = \mathrm{i}\sqrt{\frac{2\pi}{3}} r(Y_{1,-1} + Y_{1,1})\,,\ r_z = \sqrt{\frac{4\pi}{3}} r Y_{1,0}\,. \tag{II.41}$$

With this, the oscillator strength becomes

$$f_{\mathrm{i}\to\mathrm{f}} = \frac{4\pi}{3}\frac{2}{3}\bar{E}_{\mathrm{f,i}}\langle rR_\mathrm{f}, r^2/a_\mathrm{B} rR_\mathrm{i}\rangle^2 \sum_{m_j'=-j'}^{j'} \sum_{q=-1}^{1} |\langle \boldsymbol{\Phi}_\mathrm{f}, Y_{1,q}\boldsymbol{\Phi}_\mathrm{i}\rangle|^2 \tag{II.42}$$

Employing relations from [7, p. 236, 244, 260, 445, 496] one can show that the remaining matrix element is

$$\langle \boldsymbol{\Phi}_\mathrm{f}, Y_{1,q}\boldsymbol{\Phi}_\mathrm{i}\rangle = (-1)^{j'+j+1+l'+s-m_j'}\sqrt{\frac{3}{4\pi}} \begin{pmatrix} j' & 1 & j \\ -m_j' & q & m_j \end{pmatrix} \sqrt{(2j+1)(2j'+1)(2l+1)} \begin{Bmatrix} l & s & j \\ j' & 1 & l' \end{Bmatrix} C^{l,0,1,0}_{l',0}\,, \tag{II.43}$$

expressed in terms of a Wigner $3jm$-symbol, a Wigner $6j$-symbol, and a Clebsch-Gordan coefficient. With symmetry relations and a sum rule from [7, p. 236, 259] we can compute the sum over q and m_j' in the expression for the oscillator strength, which yields

$$\sum_{m_j'} f_{\mathrm{i}\to\mathrm{f}} = \frac{2}{3}\bar{E}_{\mathrm{f,i}} |\langle rR_{n',l',s,j'}, r^2/a_\mathrm{B} R_{n,l,s,j}\rangle|^2 a_l\,,\ a_l = l_\mathrm{max}(2j'+1) \begin{Bmatrix} l & s & j \\ j' & 1 & l' \end{Bmatrix}^2\,, \tag{II.44}$$

where $l_\mathrm{max} = \max(l, l')$. There are only six transitions for which a_l is non-zero. Using relations from [7, p. 298, 300] we arrive at the results shown in Tab. 1. With this it is now straight-forward to compute the oscillator strengths numerically with a computer program. As a final remark, note that the oscillator strength does not depend on m_j any more, i.e., by summing over final states, we find that the oscillators for different m_j all have the same strength.

(l, j)	$\mapsto$	(l', j')	a_l
$(l, l+s)$	$\mapsto$	$(l+1, l+s)$	$\frac{1}{(2l+3)(2l+1)}$
$(l, l-s)$	$\mapsto$	$(l+1, l+s)$	$\frac{l+1}{2l+1}$
$(l, l+s)$	$\mapsto$	$(l+1, l+3s)$	$\frac{l+2}{2l+3}$
$(l, l-s)$	$\mapsto$	$(l-1, l-s)$	$\frac{1}{(2l+1)(2l-1)}$
$(l, l+s)$	$\mapsto$	$(l-1, l-s)$	$\frac{l}{2l+1}$
$(l, l-s)$	$\mapsto$	$(l-1, l-3s)$	$\frac{l-1}{2l-1}$

Table 1: Angular factors. Table containing the angular factors a_l [see Eq. (II.44)] for computing oscillator strengths. Throughout this table, $s = 1/2$.

v. State lifetimes

To compute the lifetime of a fine-structure state, we need to calculate the so-called 'Einstein-A' coefficient [5, 8]. It is a rate that describes the number of spontaneous emissions per unit time at zero temperature. The total decay rate, at zero temperature, is given by the sum over all Einstein-A coefficients. In the dipole approximation this sum is given by the expression [5]:

$$A_{\mathrm{i}} = \sum_{\mathrm{f}} A_{\mathrm{i}\to\mathrm{f}}\,,\ A_{\mathrm{i}\to\mathrm{f}} = \frac{E_{\mathrm{H}}}{\hbar}\frac{4}{3}\alpha^2 \bar{E}^3_{\mathrm{i,f}} \sum_{u=x,y,z} |\langle \boldsymbol{\Psi}_{\mathrm{f}}, r_u/a_{\mathrm{B}} \boldsymbol{\Psi}_{\mathrm{i}}\rangle|^2\,. \tag{II.45}$$

In terms of the oscillator strength we obtain

$$A_{\mathrm{i}} = \frac{2\alpha^3 E_{\mathrm{H}}}{\hbar} \sum_{n',l',j'} \bar{E}^2_{\mathrm{i,f}}(-f_{\mathrm{i}\to\mathrm{f}})\,. \tag{II.46}$$

The rate in Eq. (II.46) is the spontaneous emission rate, which only appears for transitions where $E_{\mathrm{i}} > E_{\mathrm{f}}$. Note that, implicitly, this restricts the quantum numbers the sum is running over. Further, this is the correct decay rate only for zero temperature case. At finite temperature, spontaneous transitions increasing the energy may be driven as well, if thermal fluctuations are sufficiently strong. The background spectral density is given by Planck's law, which yields for the average photon number at frequency ω and temperature T:

$$n(\omega, T) = \frac{1}{\exp(\hbar\omega/(k_{\mathrm{B}}T)) - 1}\,, \tag{II.47}$$

where k_{B} is Boltzmann's constant. From Einstein's rate equations [8] it follows that the total rate for decay out of the initial state is

$$\frac{\hbar}{E_{\mathrm{H}}}\Gamma = 2\alpha^3 \sum_{n',l',j'} \bar{E}^2_{\mathrm{i,f}}(-f_{\mathrm{i}\to\mathrm{f}})(1 + n(E_{\mathrm{i,f}}/\hbar, T)) + 2\alpha^3 \sum_{n',l',j'} \bar{E}^2_{\mathrm{f,i}} f_{\mathrm{i}\to\mathrm{f}}\, n(E_{\mathrm{f,i}}/\hbar, T)\,, \tag{II.48}$$

where each sum only runs over quantum numbers such that the associated term is positive. That is, the first sum is only over final states with lower, and the second one only over final states with larger, energy than the initial state. With this formula, the lifetime of a fine-structure state is $\tau = \Gamma^{-1}$.

III. Manual

Reference manual for the `AlkCalc` library functions. Prototypes for the following functions are defined in the header 'interface/alkcalc.h'.

There is always a decision to be made when dealing with half-integer quantum numbers numerically. Here we agree on the following: say q is a quantum number that may take half-integer values, e.g., j, m_j. To 'clean up' the numerical value we do $2q \mapsto \lfloor 2.0 \times q + 0.5 \rfloor$. The 'floor' function is defined as $\lfloor x \rfloor = k \in \mathbb{Z}$ ($x \in \mathbb{R}$), where k is the unique integer such that $x \in [k, k+1)$. In practice, this means that for any $q \in [-0.75, -0.25)$ we use $q = -1/2$, for $q \in [-0.25, 0.25)$ we use $q = 0$, for $q \in [0.25, 0.75)$ we use $q = 1/2$, and so on, in both the negative and positive direction. In the following manual, this convention is applied to all quantum numbers that can take half-integer values, unless stated otherwise.

FUNCTION

`alkcalc_Enlsj`

DESCRIPTION

Eigenenergies in units of Hartree. The requested eigenenergy is read from the corresponding data file in the directory 'data' and returned.

ARGUMENTS

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double E`

E is the eigenenergy specified by the quantum numbers n, l, $s = 1/2$, and j.

FUNCTION

`alkcalc_fnlsj`

DESCRIPTION

Numerical equivalent to $f_{n,l,s,j}$ in Eq. (II.21). The data is read from the file containing the requested state. This data is stored at a user specified location. The location where `AlkCalc` will search for the data can be set in the file 'interface/alkcalc.h'. The result is owned by the caller and should be freed with `alkcalc_state_free`.

ARGUMENTS

`char result`

Character for deciding if discretisation data is returned; see RETURN down below. If `result` is `f` (full), the full result is returned, and if `result` is `p` (partial), the discretisation data is not returned.

`char *species`

String to specify atom/ion species, e.g., `1H` for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`alkcalc_state *f`

`f` contains eight members: `N`, `dim`, `t`, `h`, and `fnlsj`.

- `int32_t N`: Number of points, N, to discretise the interval $[0, t_{\max}]$ (see Sec. II).
- `int32_t n`: Principal quantum number.
- `int32_t l`: Orbital angular momentum quantum number.
- `double j`: Total angular momentum quantum number.
- `int32_t dim`: Dimension of eigenproblem, i.e., `dim` $= N - 2$ [see Eq. (II.30)].
- `double *t`: Array of length N, containing the mesh $(t_k)_{k=0,\ldots,N-1}$ (see Sec. II).
- `double *h`: Array of length $N-1$, containing the steps $(h_k)_{k=1,\ldots,N-1}$ (see Sec. II).
- `double *fnlsj`: `dim`-dimensional eigenvector f [see Eq. (II.30)].

FUNCTION

alkcalc_rp

DESCRIPTION

Radial matrix element of r^p, $p \in \mathbb{R}$ [see Eq. (II.36)]. For $|p| \gtrsim 10$ there is the risk of numerical over or underflow of the 64-bit floating-point arithmetic. It is the user's responsibility to ensure that the integral associated to the desired matrix element exists. In case it does not, the returned value is wrong and no error is thrown. ^{1}H (Hydrogen atom) and ^{2}He^{+} (Helium ion) can be used to benchmark atoms and ions, respectively.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the ^{88}Sr^{+} ion.

`int32_t nb`
Principal quantum number of bra.

`int32_t lb`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of bra.

`double sb`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jb`
Total angular momentum quantum number of bra, j.

`double p`
Power of radius operator; cf. Eq. (II.36). For instance, $p = 0$ yields the scalar product of the states. For $p = 1$ and $p = 2$, respectively, dipole and quadrupole matrix elements are computed. Generally, $p \in \mathbb{R}$.

`int32_t nk`
Principal quantum number of ket.

`int32_t lk`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of ket.

`double sk`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jk`
Total angular momentum quantum number of ket, j.

RETURN

`double rp`
$r_p^{n,l,s,j;n',l',s',j'}$ in Eq. (II.36) in units of a_{B}^p, where a_{B} is Bohr's radius. Quantum numbers with a prime correspond to the bra, and the ones without one, to the ket.

FUNCTION

`alkcalc_cj1m1j2m2jmj`

DESCRIPTION

Clebsch-Gordan coefficients, $C^{j_1,m_1,j_2,m_2}_{j,m_j} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m_j \rangle$, where j is the total angular momentum composed of the two individual ones j_1 and j_2. To fix the phase uniquely, we adopt the Condon-Shortley sign convention, that is, the Clebsch-Gordan coefficients with $m_j = j_1 + j_2$ are positive. In particular, for the case $j_1 = l$ and $j_2 = s$,

$$C^{l,m_l,s,m_s}_{j,m_j} = \left\langle Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}, \Phi_{l,s,j,m_j} \right\rangle , \tag{III.1}$$

with the states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$ of the uncoupled basis and the coupled basis states $\Phi_{l,s,j,m}$; see Eqs. (II.12, II.16), and Eq. (II.17), respectively. The return type (see RETURN down below) allows to extract the requested coefficient symbolically as well as numerically. The function returns zero in case the arguments do not make sense, e.g., if $j_1 < m_1$; or if j_1 is half-integer and m_1 is integer; or if j does not lay between (including bounds) $|j_1 - j_2|$ and $|j_1 + j_2|$.

ARGUMENTS

`double j1`
: Angular momentum quantum number of first factor.

`double m1`
: Magnetic quantum number of first factor.

`double j2`
: Angular momentum quantum number of second factor.

`double m2`
: Magnetic quantum number of second factor.

`double j`
: Total angular momentum quantum number.

`double mj`
: Magnetic quantum number of total angular momentum.

RETURN

`alkcalc_cg c`

c contains three members: `sign`, `numerator`, and `denominator`.

- `int8_t sign`: Sign, σ, of the Clebsch-Gordan coefficient.
- `int64_t numerator`: Numerator, ν, of the Clebsch-Gordan coefficient.
- `int64_t denominator`: Denominator, μ, of the Clebsch-Gordan coefficient.

The requested Clebsch-Gordan coefficient is given by $\sigma\sqrt{\nu/\mu}$. Euklid's algrithm is employed to reduce ν and μ. This representation allows to obtain Clebsch-Gordan coefficients symbolically as well as numerically.

FUNCTION

`alkcalc_YlmlXsms`

DESCRIPTION

Angular factor of the full state in the uncoupled basis, i.e., the two-component spinor $Y_{l,m_l}(\vartheta, \varphi)\boldsymbol{\chi}_{s,ms}$. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0, \pi]$ and $[0, 2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed.

ARGUMENTS

`int32_t l`
: Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`int32_t ml`
: Magnetic quantum number of orbital angular momentum.

`double s`
: Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ms`
: Magnetic quantum number of electron spin, m_s. For any $m_s < 0$ the value $m_s = -1/2$ is used internally, and for $m_s \geq 0$ the value $m_s = 1/2$.

`double theta`
: Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`
: Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`
: `s` has two components: `u` and `v`.
  `u` is the spin-up ($m_s = 1/2$) component.
  `d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_Philsjmj`

DESCRIPTION

Angular factor of the full eigenstate in the coupled (fine-structure) basis, i.e., the two-component spinor $\Phi_{l,s,j,mj}(\vartheta, \varphi)$ [see Eq. (II.17)]. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0, \pi]$ and $[0, 2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed. To compute the resulting two-component spinor, Clebsch-Gordon coefficients are needed, for which the sign convention detailed in Eq. (III.1) is employed.

ARGUMENTS

`int32_t l`
: Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$.

`double s`
: Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`
: Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

`double mj`
: Magnetic quantum number of total angular momentum, m_j.

`double theta`
: Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`
: Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`
: `s` has two components: `u` and `v`.
  - `u` is the spin-up ($m_s = 1/2$) component.
  - `d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_fitof`

DESCRIPTION

Oscillator strength for the transition between an initial state $\boldsymbol{\Psi}_\mathrm{i}$ and a final state $\boldsymbol{\Psi}_\mathrm{f}$, as defined in Subsec. iv, that is, it was already summed over the magnetic quantum number of the final state.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t ni`
Principal quantum number of the initial state $\boldsymbol{\Psi}_\mathrm{i}$.

`int32_t li`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the initial state $\boldsymbol{\Psi}_\mathrm{i}$.

`double si`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ji`
Total angular momentum quantum number of the initial state $\boldsymbol{\Psi}_\mathrm{i}$, j.

`int32_t nf`
Principal quantum number of the final state $\boldsymbol{\Psi}_\mathrm{f}$.

`int32_t lf`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the final state $\boldsymbol{\Psi}_\mathrm{f}$.

`double sf`
Electron spin quantum number of the final state $\boldsymbol{\Psi}_\mathrm{f}$. Not an argument here, since $s = 1/2$ is always assumed.

`double jf`
Total angular momentum quantum number of the final state $\boldsymbol{\Psi}_\mathrm{f}$, j.

RETURN

`double fitof`
$f_{\mathrm{i}\to\mathrm{f}}$ is the oscillator strength of the selected electronic transition, summed over m_j' of the final state. Note that because of this sum, it is the same for all m_j of the initial state.

FUNCTION

`alkcalc_tau`

DESCRIPTION

Lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} in nanoseconds. Note that, since the oscillator-strengths [see Eq. (II.44)] do not depend on m_j, also the lifetimes do not depend on it [see Eq. (II.48)].

ARGUMENTS

`double T`

Temperature of black-body excitation spectrum in Kelvin.

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t dn`

Range for principal quantum number. At zero temperature, this argument is not used. For finite temperature, the absorption processes in Eq. (II.48) allow 'decay' to states with higher energy. In principle, one needs to take into account all transitions to these states, of which there are infinitely many. Here, we only take into account `dn` levels above Ψ_{n,l,s,j,m_j} to truncate the series, for both $l' = l \pm 1$. In other words, the sum in Eq. (II.48), taking into account absorption, contains $2 \times$ `dn` terms for $l > 0$ and `dn` for $l = 0$.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double tau`

τ is the lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} (independent of m_j) in nanoseconds.

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